

Swapn Kumar Sarkar
Python Developer | Generative AI Engineer
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PROFILE SUMMARY

Generative AI Engineer and Python Developer with hands-on experience in designing and developing Retrieval-Augmented Generation (RAG) applications using Python, LangChain, Hugging Face, FastAPI, and Milvus. Experienced in building document ingestion pipelines, semantic search systems, vector database solutions, AI-powered chatbots, and integrating Large Language Models through APIs. Strong foundation in backend development, SQL, Object-Oriented Programming, REST APIs, and software engineering best practices. Passionate about building scalable AI applications that solve real-world business problems.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Generative AI & LLM Technologies:

Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), LangChain, Hugging Face Transformers, Embeddings, Milvus Vector Database, Semantic Search, Vector Search, LLM API Integration, AI Application Development

Python Development:

Core Python, Object-Oriented Programming (OOP), Data Structures, Functions, Modules & Packages, Iterators, Generators, Decorators, Lambda Functions, Exception Handling, File Handling, Multithreading, Logging, Code Optimization

Python Libraries & Data Processing:

NumPy, Pandas, Matplotlib, Data Processing, Data Analysis, Data Visualization

Database Management & SQL:

PostgreSQL, SQL, Database Design, DDL, DML, CRUD Operations, Joins, Subqueries, Common Table Expressions (CTEs), Window Functions, Indexing, Views, Stored Procedures, Triggers, Transactions

Data Structures & Algorithms:

Arrays, Stack, Queue, Linked List, Searching Algorithms, Sorting Algorithms, Algorithmic Problem Solving

Version Control & Development Tools:

Git, GitHub, Repository Management, Branching, Merging, Version Control Workflows

Additional Skills:

Machine Learning Fundamentals, NLP Basics, RAG Pipeline Development, LLM Application Deployment

EXPERIENCE

Generative AI Engineer, Selfstarter (June 2024 – Present, WB)

Roles & Responsibilities:

- Developed a Retrieval-Augmented Generation (RAG) application using LangChain, Milvus, and LLM APIs, enabling semantic search over enterprise documents and reducing manual document lookup time.
- Built document ingestion pipelines for processing PDFs, Word documents, web content, structured data sources.
- Implemented semantic search solutions using embedding and vector database Milvus.
- Designed and optimized chunking, embedding and retrieval strategies to enhance search relevance and reduce LLM hallucinations.
- Developed AI-powered Chatbot.
- Integrated Large Language Models through APIs for content generation, summarization.

Projects:

PDF RAG Chatbot link: <https://ragpipe.streamlit.app>

Upgrade Demo link:

https://drive.google.com/file/d/1AxyOCxTE2HMCko5kv8sZ-_7WKkg-qmHY/view?usp=drivesdk

Features:

- PDF upload and document parsing
- Automatic document chunking
- Embedding generation
- Vector search using Milvus
- Context-aware question answering
- Interactive Chatbot interface

Sr. Network Engineer, Nipun Net Solution Pvt Ltd (June 2023 – June 2024, Hyderabad)

Roles & Responsibilities:

- Configured and maintained Cisco 2960 and Cisco 3850 enterprise switches.
- Implemented VLANs, LACP, HSRP, DHCP, IP Routing, and Trunking.
- Configured Aruba 2930 switches.
- Performed switch backup, maintenance, and troubleshooting.
- Managed enterprise network infrastructure and user connectivity.

Customer Support Engineer, Informatics Technologies Pvt Ltd (Aug2022 – May2023, Ahmedabad)

Roles & Responsibilities:

- Developed backend functionality using Python
- Integrated PostgreSQL database
- Built reusable and maintainable code
- Tested and debugged application features

Technologies: Python, PostgreSQL

Desktop Support Engineer, IMSI Staffing(A TeamLease Company) (Feb2021 – Aug2022 , Haldia,WB)

Roles & Responsibilities:

- Managed Windows Server administration.
- Configured Active Directory, DNS, DHCP, Group Policy, and Office 365 integration.
- Performed desktop, server, and network troubleshooting.
- Managed backup and recovery processes.

Associate System Engineer, Precision Techserve (M) Pvt. Ltd (Oct2019 – Sep2020 , Hooghly,WB)

Roles & Responsibilities:

- Installed and configured Windows operating systems.
- Managed Active Directory users and policies.
- Provided L1/L2 desktop support.
- Resolved network connectivity issues.

Software Support Engineer, Aidni Infotech Pvt Ltd (April 2019 – Sep2019 , Kolkata)

Roles & Responsibilities:

- Provided software support
- Assisted clients in requirement gathering
- Supported backend and frontend web applications

QUALIFICATION

- B.Sc. in Mathematics, University of Calcutta, 2014
- Higher Secondary (Science), W.B.C.H.S.E, 2011

TECHNICAL QUALIFICATION

Post Graduate Diploma in Software Engineering, from THE INSTITUTE OF COMPUTER ENGINEERS

[INDIA] (not appeared in exam)

Completed a Networking Course CCNA on Routing & Switching from IINT During Mar- 2017 to Feb-2018

SOFT SKILLS

- Problem-Solving
- Communication
- Collaboration
- Time Management
- Critical Thinking
- Adaptability

PERSONAL DETAILS

Languages Known: Bengali, English, Hindi

Nationality: Indian

Marital Status: Unmarried

Gender: Male

Category: General